Research Synopsis

My interest in artificial intelligence comes from a deeper curiosity about how intelligent systems can assist humans in solving complex real world problems. Over time, this interest gradually evolved toward healthcare focused AI, where machine learning systems have the potential to support critical decision making, improve patient outcomes, and make healthcare systems more efficient and accessible. Among the research themes offered at Jio Institute, I am particularly interested in Advanced AI for Healthcare and Scientific Systems under the guidance of Dr. Sudipta Roy because it strongly aligns with both my academic interests and practical project experience.

One of the major challenges in healthcare AI today is the development of systems that are not only accurate but also reliable, interpretable, and practical in real clinical environments. While many machine learning models can achieve strong predictive performance, healthcare professionals often hesitate to trust them fully because the reasoning behind predictions is unclear. This challenge becomes even more important in high risk medical scenarios where decisions directly affect patient safety. I am especially interested in exploring how advanced AI systems can combine predictive capability with explainability and intelligent decision support to create systems that healthcare professionals can trust and effectively use.

My interest in this area deepened significantly while working on my healthcare AI project called PranRakshak AI. The project focuses on early sepsis risk prediction, patient prioritization, and intelligent clinical decision support. Sepsis is a serious medical condition where early detection can greatly improve survival rates, yet identifying it early remains difficult because symptoms often overlap with other conditions. Through this project, I explored how machine learning models can be applied to healthcare data for early risk prediction while also integrating explainable AI techniques to improve transparency and interpretability.

Working on PranRakshak AI gave me valuable exposure to the complete pipeline involved in building applied AI systems. I worked on healthcare data preprocessing, feature engineering, machine learning model development, prediction pipelines, and full stack deployment. I also implemented SHAP based explainability methods to better understand model behavior and provide interpretable outputs for decision support. Beyond the technical aspects, the project helped me understand the practical limitations and ethical considerations involved in healthcare AI systems. It strengthened my interest in developing AI systems that are not only technically strong but also responsible and human centered.

Apart from healthcare AI, I have also explored broader areas of artificial intelligence through projects involving natural language processing, retrieval augmented generation systems, and predictive modeling. My project RAGify introduced me to concepts such as embeddings, vector databases, retrieval pipelines, and large language model integration. These experiences improved my understanding of modern AI architectures and strengthened my ability to learn and adapt to new technologies and research directions. I have also worked on NLP based projects such as Quora Question Pair similarity detection, which further improved my understanding of machine learning workflows and model evaluation.

Academically, my undergraduate studies in Computer Science and Engineering at I.K. Gujral Punjab Technical University have provided me with a strong foundation in mathematics, programming, machine learning, and data structures. Alongside formal coursework, I actively pursue self driven learning through research papers, online courses, and practical implementation based projects. I enjoy approaching problems analytically and systematically, and I believe this mindset is important for meaningful research.

I am particularly drawn toward research environments where learning is driven by curiosity, experimentation, and critical thinking. I believe that the opportunity to work under experienced researchers at Jio Institute would help me deepen my understanding of advanced AI systems and scientific research methodologies. More importantly, I hope to contribute actively through my technical skills, project experience, and willingness to learn.

In the long term, I aim to pursue advanced research and contribute to the development of impactful AI systems that solve meaningful real world problems. I view this internship as an important step toward that goal because it would allow me to gain deeper exposure to research oriented problem solving, collaborate with researchers, and strengthen my understanding of advanced AI applications in healthcare and scientific systems.

Overall, I believe that my background in machine learning, healthcare AI projects, natural language processing, and full stack AI development aligns strongly with this research theme. My curiosity for understanding intelligent systems, combined with my practical project experience and research oriented mindset, motivates me to pursue this opportunity with sincerity and enthusiasm.